WORKED SOLUTION

STEP III 2016, Question 11

Mechanics • Power and Resistance

STEP III 2016, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

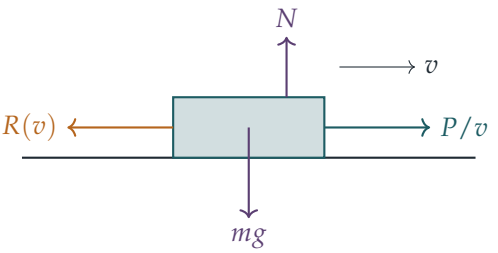
step.maths.org/questions/16-s3-q11

The key idea

An engine working at constant rate P exerts a driving force P/v at speed v. Since we want *distance* as a function of speed, not time, we write the acceleration as $v\frac{\mathrm{d}v}{\mathrm{d}x}$. That turns Newton's second law into a separable equation for x in terms of v. The fact that the acceleration vanishes at $4U$ fixes the resistance constant in terms of P and U, and each distance then comes down to a single definite integral.

Solution

We model the car as a particle of mass m moving along a horizontal line, with tractive force P/v forwards and resistance $R(v)$ backwards. Vertically the weight and normal reaction balance and play no further part.



Using $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$, the horizontal equation of motion is

$$mv\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{P}{v} - R(v), \qquad \text{so} \qquad \frac{\mathrm{d}x}{\mathrm{d}v} = \frac{mv^2}{P - vR(v)}.$$

In both models $vR(v)$ increases with v and equals P at $v = 4U$. So for $U \le v \le 2U$ the denominator is positive, and the car really is accelerating throughout, which means x increases with v and the distance is

$$X = \int_U^{2U} \frac{mv^2}{P - vR(v)}\,\mathrm{d}v.$$

Part (i)

Let $R = kv$. Zero acceleration at $4U$ gives $\dfrac{P}{4U} = 4kU$, so $k = \dfrac{P}{16U^2}$. Then

$$P - kv^2 = k\,(16U^2 - v^2), \qquad \frac{m}{k} = \frac{16mU^2}{P} = \frac{1}{\lambda U}.$$

Hence

$$\lambda X_1 = \frac{1}{U}\int_U^{2U} \frac{v^2}{16U^2 - v^2}\,\mathrm{d}v = \int_1^2 \frac{w^2}{16 - w^2}\,\mathrm{d}w,$$

substituting $v = Uw$. Writing $\dfrac{w^2}{16 - w^2} = -1 + \dfrac{16}{16 - w^2} = -1 + \dfrac{2}{4 - w} + \dfrac{2}{4 + w}$,

$$\begin{aligned}\lambda X_1 &= \Big[-w + 2\ln\frac{4 + w}{4 - w}\Big]_1^2 \\ &= -1 + 2\ln 3 - 2\ln\tfrac{5}{3} = 2\ln\tfrac{9}{5} - 1,\end{aligned}$$

as required.

Part (ii)

Now let $R = cv^2$. Zero acceleration at $4U$ gives $\dfrac{P}{4U} = 16cU^2$, so $c = \dfrac{P}{64U^3}$. Then

$$P - cv^3 = c\,(64U^3 - v^3), \qquad \frac{m}{c} = \frac{64mU^3}{P} = \frac{4}{\lambda}.$$

Ask yourself Is there a neat antiderivative here, or do we need partial fractions again?

The numerator v^2 is, up to a constant, the derivative of the denominator $64U^3 - v^3$. So this one is a logarithm straight away:

$$\begin{aligned}\lambda X_2 &= 4\int_U^{2U} \frac{v^2}{64U^3 - v^3}\,\mathrm{d}v = 4\Big[-\tfrac{1}{3}\ln(64U^3 - v^3)\Big]_U^{2U} \\ &= \tfrac{4}{3}\ln\frac{63U^3}{56U^3} = \tfrac{4}{3}\ln\tfrac{9}{8},\end{aligned}$$

as required.

Part (iii)

Since $\lambda > 0$, $X_1 > X_2$ exactly when $\lambda X_1 - \lambda X_2 > 0$. We only have bounds for $\ln 24$ and $\ln 5$, so we rewrite the difference in terms of those:

$$\begin{aligned}\lambda(X_1 - X_2) &= 2\ln 9 - 2\ln 5 - 1 - \tfrac{4}{3}\ln 9 + \tfrac{4}{3}\ln 8 \\ &= \tfrac{4}{3}\ln 3 + 4\ln 2 - 2\ln 5 - 1 \\ &= \tfrac{4}{3}(\ln 3 + 3\ln 2) - 2\ln 5 - 1 = \tfrac{4}{3}\ln 24 - 2\ln 5 - 1.\end{aligned}$$

Ask yourself Which way round should we use each bound?

To show the difference is positive we need a *lower* bound, so we use the lower bound for $\ln 24$ (which appears with a plus sign) and the upper bound for $\ln 5$ (which appears with a minus sign):

$$\lambda(X_1 - X_2) > \tfrac{4}{3}(3.17) - 2(1.61) - 1 = 4.22\dot{6} - 3.22 - 1 = 0.00\dot{6} > 0.$$

Therefore $X_1 > X_2$: the car covers more ground getting from U to $2U$ when the resistance is proportional to speed than when it is proportional to speed squared.

This makes physical sense. Both resistances agree at $v = 4U$, but for $v < 4U$ we have $cv^2 = kv \cdot \frac{v}{4U} < kv$. So the squared law resists *less* at the speeds in question, the car accelerates harder, and it reaches $2U$ sooner.

Checks

- Numerically, $\lambda X_1 = 2\ln 1.8 - 1 \approx 0.1756$ and $\lambda X_2 = \frac{4}{3}\ln 1.125 \approx 0.1570$, so the difference is about 0.0185, which is positive, as found. A computer simulation stepping the equation of motion in time from $v = U$ reproduces both values to five decimal places.
- Units: $P/(mU^3)$ has units $\mathrm{W}/(\mathrm{kg\,m^3\,s^{-3}}) = \mathrm{m^{-1}}$, so λX is dimensionless, as it should be.
- The physical argument at the end of part (iii) predicts the same ordering independently.

Common slips

- Using $a = \frac{\mathrm{d}v}{\mathrm{d}t}$, which gives time rather than distance. Here we need $v\,\frac{\mathrm{d}v}{\mathrm{d}x}$.
- Using the bounds the wrong way round in part (iii). Using 3.18 and 1.60 only gives an upper bound, which proves nothing about the sign.
- Forgetting that the margin is tiny ($0.00\dot{6}$), so rounding $\frac{4}{3}(3.17)$ to 4.2 loses the result.

Beyond STEP

Writing $a = v\,\mathrm{d}v/\mathrm{d}x$ is the first step towards the energy method of university mechanics. Multiplying the equation of motion by v gives the rate of change of kinetic energy, which equals power in minus power dissipated. Speeds like $4U$, where the two balance, are *equilibria* of a first-order differential equation, and the study of how solutions approach them is the beginning of dynamical systems. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats work, power and velocity-dependent forces in this style.

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